

Nazmul Alam Diptu

+880-1845511889 | diptunazmulalam@gmail.com | [linkedin.com/in/diptu](https://www.linkedin.com/in/diptu) | [diptu.github.io](https://github.com/diptu) | Dhaka, Bangladesh

EXPERIENCE

North South University

May 2019 – May 2025

Lab Instructor

Dhaka, Bangladesh

- Delivered 6 years of lab instruction in **C, Java, Data Structures, Databases, and Web Technologies**; mentored 100+ undergraduates in algorithmic problem-solving and research methodology
- Collaborated with faculty on **curriculum design and student assessment**, contributing directly to program-level academic outcomes

Me-Solshare Ltd

Nov. 2019 – Aug. 2021

Jr. Data Scientist

Dhaka, Bangladesh

- Built **CNN-based time-series models achieving 95% accuracy** for energy forecasting; applied statistical modeling and experimental evaluation aligned with graduate-level ML research practice

Power Ledger Pty Ltd

Oct. 2022 – Feb. 2024

Product Analyst

Perth, Australia (Remote)

- Performed **data analysis, system QA, and technical reporting** for a blockchain energy trading platform; bridged engineering and product teams across international time zones

PUBLICATIONS & PROJECTS

End-to-End OCR via Synthetic Dataset Generation (*IJCCI 2020, Springer*)

- Trained a **deep-learning OCR model** on synthetically generated noisy images; improved robustness in low-quality document recognition | [doi:10.1007/978-981-15-3607-6_41](https://doi.org/10.1007/978-981-15-3607-6_41)

Population Estimation of Rohingya Refugees via ANN (*VJCS 2019, World Scientific*)

- Modeled refugee population using **Artificial Neural Networks** on satellite and census data; applied to humanitarian resource planning | [doi:10.1142/S2196888819500246](https://doi.org/10.1142/S2196888819500246)

CarbonIQ — AI-Powered Carbon Accounting Platform

- Designed **time-series CO₂e forecasting pipelines** and automated ETL/ELT workflows; built multi-tenant API architecture with containerized, reproducible MLOps infrastructure

EDUCATION

North South University

Jan. 2015 – Dec. 2018

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

- CGPA: 3.62 / 4.0** | **Coursework:** Machine Learning, Algorithms, Databases, Computer Systems, OOP
- Certifications:** Deep Learning Specialization (DeepLearning.AI, 2022); Machine Learning Specialization (Stanford/Coursera, 2021)

SKILLS

Languages : Python, C, C++, JavaScript/TypeScript, Java, SQL

ML & Research : PyTorch, TensorFlow, scikit-learn, MLflow, Airflow, dbt, Pandas, NumPy, Matplotlib

Tools & Cloud : FastAPI, Docker, PostgreSQL, MongoDB, GitHub Actions, AWS, Linux, $\LaTeX$